

in history – the blue ball of the Earth rising above the bleak horizon of the Moon, set against the darkness of space. This image is widely credited with helping to create a new vision of the Earth as a beautiful but fragile planet, whose capacity to support life amid the inanimate vastness of space was a miracle that humankind should treat with much greater respect.



SATURN V LIFTOFF

A frame-by-frame record of the launch of Apollo 11 gives a vivid impression of Saturn's power. The first stage engines used up 40,000 gallons (150,000 liters) of propellant a minute.

One-horse race

After the triumph of Apollo 8, there was only one contestant left in the Moon race. A Soviet circumlunar flight would now be pointless, while the chances of a cosmonaut being first to set foot on the Moon shrank to zero as test launches of the N-1 booster repeatedly failed. The Soviets continued with some spectacular Soyuz space rendezvous and docking exercises in Earth orbit, while beginning to hint that they had never really wanted to go to the Moon at all.

Bursting with optimism, NASA now had a manned Moon landing in its sights. In March 1969, the Apollo 9 mission, crewed by James McDivitt, David Scott, and Russell Schweickart, successfully put the lunar module through its paces in Earth orbit. The following May, Apollo 10 repeated these exercises in lunar orbit. Mission commander Thomas Stafford and pilot Eugene Cernan flew the lunar module to an altitude of 50,000ft (15,000m) above the surface of the Moon while their colleague John Young awaited their return on board the orbiting command module. Once again, the trip to the Moon and back was accomplished without drama.

ONE SMALL STEP

Ever aware of the importance of television coverage, NASA had arranged for a camera to film Neil Armstrong emerging from the Eagle landing module and climbing down the ladder onto the Moon's surface.

Under the stress of the occasion Armstrong slightly stuffed the lines that he had been mentally rehearsing, coming out with: "That's one small step for man, one giant leap for mankind."



Big moment approaches

All was now ready for a manned landing on the Moon. Apollo 11 would be the mission to achieve the goal that President Kennedy had laid down, with more than five months to spare before the end-of-decade deadline. The crew would consist of Neil Armstrong, the mission commander; Edwin "Buzz" Aldrin; and Michael Collins. Apollo crews were assigned on a strict rotation basis to minimize possible conflict between a notoriously competitive set of individuals. That Armstrong's team landed the mission was therefore pure chance, not a reflection of any

